

40.520 Stochastic Models

Continuous Time Markov Chain

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Outline

1. Introducing Continuous -Time Markov Chain
2. Kolmogorov Backward/Forward Equations
3. Limiting Probabilities
 - Embedded Markov Chain
 - Uniformization
4. Classical Queueing Examples.

Continuous-Time Markov Chain

Recap: Discrete-Time Markov Chains (DTMC)

Definition (DTMC)

A time-homogeneous DTMC (Discrete-Time Markov Chain) is a stochastic process $\{X_n, n = 0, 1, 2, \dots\}$, where X_n denotes the state at (discrete) time step n and such that, $\forall n \geq 0$, $\forall i, j$, and $\forall i_0, \dots, i_{n-1}$:

$$\begin{aligned} & \mathbf{P}\{X_{n+1} = j \mid X_n = i, X_{n-1} = i_{n-1}, \dots, X_0 = i_0\} \\ &= \mathbf{P}\{X_{n+1} = j \mid X_n = i\} \\ &= P_{ij}. \end{aligned}$$

Three Properties of A DTMC

1. Transitions are made at discrete time steps, $n = 0, 1, 2, \dots$
2. The past does not matter. Only the present state matters.
3. Transition probabilities are “stationary”.

Continuous-Time Markov Chain (CTMC)

Definition (CTMC)

A time-homogeneous Continuous-Time Markov Chain (CTMC) is a continuous-time stochastic process $\{X(t), t \geq 0\}$ such that, $\forall s, t \geq 0$ and $\forall i, j, x(u)$:

$$\begin{aligned} & \mathbf{P}\{X(t+s) = j \mid X(s) = i, X(u) = x(u), 0 \leq u < s\} \\ &= \mathbf{P}\{X(t+s) = j \mid X(s) = i\} \\ &= \mathbf{P}\{X(t) = j \mid X(0) = i\} = P_{ij}(t) \end{aligned}$$

Three Properties of A DTMC

- Transitions between states can happen at any time.
- Markovian Property: Only the present state matters.
- Transition probabilities are “stationary”.

Memoryless Holding Times

Holding Time at state i . Define τ_i to be the time until the CTMC leaves state i , given that the CTMC is currently in state i . Then

$$\mathbf{P}\{\tau_i > t + s \mid \tau_i > s\} = \mathbf{P}\{\tau_i > t\}$$

Implication

- τ_i is memoryless
- τ_i follows exponential distribution

CTMC - View 1

Definition (CTMC - View 1)

A CTMC is a stochastic process with the property that every time it enters state i , the following hold:

1. The amount of time the process spends in state i before making a transition is Exponentially distributed with rate ν_i .
2. When the process leaves state i , it will next enter state j with **jump probability** p_{ij} independent of the time spent at state i .

Remark: We have a Discrete-Time Markov Chain embedded in the process with transition probabilities being the jump probabilities p_{ij} .

Embedded Markov Chain

The **embedded Markov chain** is the discrete-time Markov chain formed by the sequence of states visited by a continuous-time Markov chain (CTMC) *at the moments of transition*.

A New Perspective on Poisson Process

- A Poisson process with rate λ can be viewed as a CTMC:
 - At each state n , it will stay for an exponential time with mean $1/\lambda$;
 - Then, it proceeds from n to $n + 1$.

An Example

- $S = \{1, 2, 3\}$
- “Jump” matrix

$$\begin{pmatrix} 0 & 2/3 & 1/3 \\ 1/2 & 0 & 1/2 \\ 0 & 1 & 0 \end{pmatrix}$$

- Holding time rates

$$v_1 = 3, \quad v_2 = 2, \quad v_3 = 1$$

CTMC - View 2

Definition (CTMC - View 2)

Let $X_j \sim \text{Exp}(\nu_i p_{ij})$ represent the time to transition from i to j , $\forall j \neq i$. Let $\tau_i = \min_j \{X_j\}$ be the time until the Markov chain leaves state i . Let the next state be m where $m = \text{argmin}_j \{X_j\}$, i.e., X_m was the $\min_j \{X_j\}$.

Interpretation

- Being in state i and waiting for multiple independent “calls” from other states
- The time until a call from state j is $X_j \sim \text{Exp}(\nu_i p_{ij})$.
- The first call received determines the next state.

Q-Matrix

The **rate matrix** or **infinitesimal generator** $\mathbf{Q} = [Q_{ij}]$ describes the transition rates of a continuous-time Markov chain.

- $Q_{ij} = q_{ij} = p_{ij}\nu_i$ for $i \neq j$
- $Q_{ii} = -\nu_i = -\sum_{j \neq i} q_{ij}$. The diagonal entries are negative and represent the *total exit rate* from state i .

$$\mathbf{Q} = \begin{pmatrix} -\nu_1 & p_{12}\nu_1 & p_{13}\nu_1 & \cdots \\ p_{21}\nu_2 & -\nu_2 & p_{23}\nu_2 & \cdots \\ p_{31}\nu_3 & p_{32}\nu_3 & -\nu_3 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

where $\sum_{j \neq i} p_{ij} = 1$.

Q-Matrix

- The **Q** matrix fully characterizes the CTMC: rates determine both holding times and jump probabilities.
- Relationship: **Q**-Matrix and transition matrix **P** of the embedded Markov chain:

$$p_{ij} = \frac{Q_{ij}}{\nu_i} \quad (i \neq j)$$

Example

Given the Q -Matrix of a CTMC:

$$\mathbf{Q} = \begin{pmatrix} -(\lambda_{12} + \lambda_{13}) & \lambda_{12} & \lambda_{13} \\ \lambda_{21} & -(\lambda_{21} + \lambda_{23}) & \lambda_{23} \\ \lambda_{31} & \lambda_{32} & -(\lambda_{31} + \lambda_{32}) \end{pmatrix}$$

find the transition matrix of the embedded Markov chain.

Kolmogorov Backward/Forward Equations

Transition Probability Function

- Let $\{X_t, t \geq 0\}$ record the state of a CTMC over time. Then

$$P_{ij}(t) = P(X_{t+s} = j \mid X_s = i)$$

is known as the transition probability function of the process, i.e., the probability of being in state j at time t , given that the process started in state i at time 0.

- We have

$$P_{ij}(t+s) = \sum_{k=1}^n P_{ik}(t)P_{kj}(s).$$

Kolmogorov Backward/Forward Equations

Theorem (Kolmogorov Backward Equation) For states i, j , and times $t \geq 0$,

$$P'_{ij}(t) = \sum_{k \neq i} q_{ik} P_{kj}(t) - \nu_i P_{ij}(t),$$

where

$$q_{ik} = \nu_i p_{ik}.$$

Kolmogorov Backward/Forward Equations

Theorem (Kolmogorov Forward Equation) For states i, j , and times $t \geq 0$,

$$P'_{ij}(t) = \sum_{k \neq j} q_{kj} P_{ik}(t) - \nu_j P_{ij}(t).$$

Example: Two-State Chain

- Consider a machine that works for an exponential amount of time having mean $1/\lambda$ before breaking down; and suppose that it takes an exponential amount of time having mean $1/\mu$ to repair the machine.
- If the machine is in working condition at time 0, then what is the probability that it will be working at time $t = 10$?

First Order Non-Homogeneous Linear ODE

Given a first order non-homogeneous linear differential equation

$$y' + p(t)y = f(t),$$

using variation of parameters the general solution is given by

$$y(t) = v(t)e^{P(t)} + Ae^{P(t)},$$

where $v'(t) = e^{-P(t)}f(t)$ and $P(t)$ is an antiderivative of $-p(t)$.

General CTMC

- For any pair of states i and j , let

$$Q_{ij} = \begin{cases} q_{ij}, & i \neq j \\ -v_i, & i = j \end{cases}.$$

- Then, we can rewrite the backward/forward equations as follows:

$$P'_{ij}(t) = \sum_k Q_{ik} P_{kj}(t) \quad (\text{backward})$$

$$P'_{ij}(t) = \sum_k P_{ik}(t) Q_{kj} \quad (\text{forward})$$

Computing Transition Probabilities

- Introduce matrices Q , $P(t)$, $P'(t)$ by letting the element in row i , column j of these matrices be, respectively, Q_{ij} , $P_{ij}(t)$, $P'_{ij}(t)$.
- The two equations can be written as:

$$P'(t) = QP(t) \quad (\text{backward})$$

$$P'(t) = P(t)Q \quad (\text{forward})$$

- The solution should be

$$P(t) = \sum_{n=0}^{+\infty} Q^n \frac{t^n}{n!}.$$

Limiting Probabilities

Limiting Probabilities

- The probability that a continuous-time Markov chain will be in state j at time t often converges to a limiting value that is independent of the initial state.
- We intend to study

$$\pi_j = \lim_{t \rightarrow +\infty} P_{ij}(t) = ?$$

Limiting Probabilities

- To derive a set of equations for the π_i , we may let t approach ∞ in the forward equation. Then,

$$0 = \sum_{k \neq j} q_{kj} \pi_k - \nu_j \pi_j.$$

- In addition,

$$\sum_i \pi_i = 1.$$

Limiting Probabilities

- It is easy to see that if the embedded Markov chain has a limiting stationary distribution

$$\pi = (\pi_1, \dots, \pi_n), \quad \text{i.e.,} \quad \pi_i = \sum_j \pi_j p_{ji},$$

then

$$\tilde{\pi}_i \propto \frac{\pi_i}{v_i}.$$

- The solution to the equations on the last slide should be

$$\tilde{\pi}_i = \frac{\pi_i / v_i}{\sum_i \pi_i / v_i}.$$

Example: A Shoe Shine Shop

- Consider a shoe shine establishment consisting two chairs — chair 1 and chair 2. A customer upon arrival goes initially to chair 1 where his shoes are cleaned and polish is applied. After this is done, the customer moves on to chair 2 where the polish is buffed.
- The service times at the two chairs are independent and exponentially distributed with respective rates μ_1 and μ_2 .
- Suppose that potential customers arrive in accordance with a Poisson process having rate λ , and that a potential customer will enter the system only if both chairs are empty.
- Define a CTMC and find the limiting probabilities.

Uniformization: Discretization of CTMC to DTMC

Consider an M/M/1 queue example: Customers arrive according to a Poisson process of rate $\lambda > 0$. There is a single server and the service times are i.i.d. exponential with parameter $\mu > 0$.

Model by flipping two coins simultaneously every δ -step:

- First coin: probability $\lambda\delta$ of “arrival”
- Second coin: probability $\mu\delta$ of “departure”

Uniformization: Discretization of CTMC to DTMC

Four Cases

- With probability $\lambda\delta(1 - \mu\delta)$: arrival and no departure
- With probability $(1 - \lambda\delta)\mu\delta$: departure and no arrival
- With probability $\lambda\delta\mu\delta$: arrival and departure
- With probability $(1 - \text{all of the above})$: no arrival and no departure

This defines a discrete-time Markov chain that transitions *every* δ time units, with transition probabilities approximated by the rates above.

Uniformization: Discretization of CTMC to DTMC

As $\delta \rightarrow 0$, we can rewrite:

$$\lambda\delta(1 - \mu\delta) \longrightarrow \lambda\delta + o(\delta)$$

$$\mu\delta(1 - \lambda\delta) \longrightarrow \mu\delta + o(\delta)$$

$$\lambda\delta\mu\delta \longrightarrow o(\delta)$$

$$(1 - \text{all of the above}) \longrightarrow 1 - \lambda\delta - \mu\delta + o(\delta)$$

Solving the DTMC and Taking the Limit

For state 0:

$$\pi_0(\lambda\delta + o(\delta)) = \pi_1(\mu\delta + o(\delta))$$

$$\pi_0 \left(\lambda + \frac{o(\delta)}{\delta} \right) = \pi_1 \left(\mu + \frac{o(\delta)}{\delta} \right)$$

$$\lim_{\delta \rightarrow 0} \pi_0 \left(\lambda + \frac{o(\delta)}{\delta} \right) = \lim_{\delta \rightarrow 0} \pi_1 \left(\mu + \frac{o(\delta)}{\delta} \right)$$

$$\pi_0 \cdot \lambda = \pi_1 \cdot \mu \quad \Rightarrow \quad \pi_1 = \frac{\lambda}{\mu} \pi_0$$

Balance Equations for the CTMC

For state 1:

$$\pi_1(\lambda\delta + o(\delta) + \mu\delta + o(\delta)) = \pi_0(\lambda\delta + o(\delta)) + \pi_2(\mu\delta + o(\delta))$$

Taking the limit yields:

$$\pi_1(\lambda + \mu) = \pi_0\lambda + \pi_2\mu$$

Substituting $\pi_1 = \frac{\lambda}{\mu}\pi_0$:

$$\frac{\lambda}{\mu}\pi_0(\lambda + \mu) = \lambda\pi_0 + \mu\pi_2 \quad \Rightarrow \quad \pi_2 = \left(\frac{\lambda}{\mu}\right)^2 \pi_0$$

Continuing, we obtain:

$$\pi_n = \left(\frac{\lambda}{\mu}\right)^n \pi_0$$

General Balance Equations for CTMCs

Balance Equations for CTMCs

For each state j :

$$\pi_j \nu_j = \sum_i \pi_i q_{ij}$$

where $q_{ij} = \nu_i \cdot p_{ij}$.

Solving CTMC

Theorem (Summary Theorem for CTMCs)

Given an irreducible CTMC, suppose there exist π_i 's such that $\forall j$,

$$\pi_j \nu_j = \sum_i \pi_i q_{ij} \quad \text{and} \quad \sum_i \pi_i = 1.$$

Then the π_i 's are the limiting probabilities for the CTMC, and the CTMC is ergodic.

Balance $\neq$ Stationary for CTMCs

- For DTMCs, balance equations are equivalent to stationary equations.
- For a CTMC, the balance equations yield the limiting probabilities, while the stationary equations are meaningless, unless we first translate to a DTMC.

Classical Queueing Examples

Example: M/M/1 Queue

A coffee shop has **one server**. Customers arrive according to a **Poisson process** at rate λ . The service time follows an **exponential distribution** with rate μ . Find:

- (a) The limiting (stationary) distribution of the number of customers in the system
- (b) The probability the system is empty
- (c) The probability there are more than 3 customers in the system
- (d) The **throughput** (long-run departure rate)
- (e) The **average queue length** (excluding the one in service)
- (f) The **average number of customers in the system**
- (g) The server **utilization**

M/M/c Queue

A customer service call center has c identical servers. Calls arrive according to a **Poisson process** at rate λ . Each server handles calls with service times following an **exponential distribution** with rate μ . Customers who arrive when all servers are busy join a single FIFO queue. Find

- (a) The stability condition and server utilization
- (b) The limiting (stationary) distribution of the number of customers in the system
- (c) The probability that an arriving customer must wait (all servers busy)
- (d) The **throughput** (long-run departure rate)
- (e) The **average queue length** (excluding customers in service)
- (f) The **average number of customers in the system**
- (g) The probability that exactly 2 servers are busy